

Decision Control Statement Problems



1

Largest of Three Numbers

Problem Statement

Write a program to input three numbers and determine which number is the largest among them.

Input Format

First line of input contains the three integers denoting the value of a, b and c.

Output Format

Largest integer in a single line.

Sample Input

5 9 2

Sample Output

9

2

Grade CalculationProblem
Statement

Write a C++ program to input marks of five subjects, calculate the percentage, and print the grade using this rule:

90% and above → A+

75% to 89% → A

60% to 74% → B

40% to 59% → C

Below 40% → Fail

Input
Format

First line contains the value of five subjects.

Output
Format

Display the grade.

Sample
Input

95 89 92 90 93

Sample
Output

A+

3

Day of the Week

Problem
Statement

Write a program to input a number (1–7) and display the corresponding day of the week.

Input
Format

First line contain the single number.

Output
Format

Day name.

Sample
Input

1

Sample
Output

Monday

4

Roots of Quadratic EquationProblem
Statement

Write a program to input coefficients of a quadratic equation ($ax^2 + bx + c = 0$) and determine the nature of its roots.

Input
Format

First line has space separated integers value.

Output
Format

Display the nature of roots.

Sample
Input

1 -3 2

Sample
Output

Real and Equal Roots

5	Electricity Bill Calculation
Problem Statement	Write a program to input the total electricity units consumed and calculate the bill according to the following rates: For first 100 units → ₹1.50 per unit For next 100 units → ₹2.00 per unit For units above 200 → ₹3.00 per unit Also add 20% surcharge on the total bill.
Input Format	First line contains the total units consumed.
Output Format	Display the final bill.
Sample Input	75
Sample Output	135

6	Type of Triangle
Problem Statement	Write a program to input three sides of a triangle and determine whether it is Equilateral, Isosceles, or Scalene.
Input Format	First line contain the three sides of a triangle.
Output Format	Display the type of triangle.
Sample Input	3 4 6
Sample Output	Scalene

7

Menu-Driven Area Program

Problem
Statement

Write a menu-driven program to calculate area based on user choice:

1. Circle
2. Rectangle
3. Triangle

Input
Format

First line is a choice (1 to 3)
Second line contain the space seprated values as per the choice.

Output
Format

Display the Area.

Sample
Input

1
7

Sample
Output

153.86

8	Menu-Driven Area Program
Problem Statement	Write a menu-driven program to calculate area based on user choice: 4. Circle 5. Rectangle 6. Triangle
Input Format	First line is a choice (1 to 3) Second line contain the space seprated values as per the choice.
Output Format	Display the Area.
Sample Input	1 7
Sample Output	153.86

9	Salary Calculation
Problem Statement	Write a program to calculate Gross Salary of an employee based on basic salary and allowances: If Basic $\leq$ ₹10,000 $\rightarrow$ HRA = 20%, DA = 80% If Basic $\leq$ ₹20,000 $\rightarrow$ HRA = 25%, DA = 90% If Basic $>$ ₹20,000 $\rightarrow$ HRA = 30%, DA = 95%
Input Format	First line contain the Basic Salary.
Output Format	Display the Gross salary.
Sample Input	15000
Sample Output	30750